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The Extended Gate Family

Six new handles beyond the protocol; the stability of the H2 gap

Ishak Khamzatovich Isaev

Independent researcher, Nalchik, Kabardino-Balkarian Republic

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Ishak Khamzatovich Isaev · Independent researcher, Nalchik, Kabardino-Balkarian Republic
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Introduction and setting

Campaign C5 answers the natural objection to C3: perhaps identity is reached not on the protocol handles N_v , q , W , α but on something else? We extend the handle space by six new dimensions: boundary twists (φ_x , φ_y) — spin structures in the sense of C8; hopping anisotropy $t_x/t_y \in \{0.8, 1.25\}$; flux shift $\delta\alpha = \pm 0.02$ about the self-dual line ($\pm 1/24$ on the torus, where the constraint $\alpha \cdot N_y \in \mathbb{Z}$ is exact); a staggered mass $m \in \{0.15, 0.35\}$ breaking the chiral structure; the vortex profile — point-like (site) versus a finite-radius core (midpoint); the charge sign pattern — neutral (protocol), all-plus, random; and the boundary type — open versus torus.

In total 35 new gates plus 2 random-sign decoys, all at the same master seed 96, the same unfolding pipeline and the same three statistics as C3 — the comparison is batch-consistent.

Method and safeguards

Every handle is implemented in the engine.hamiltonian kernel and covered by a regression control: the protocol reference gate ($N_v = 0$, $W = 0.5$, $\alpha = 0.5$) reproduces the mean level spacing of the C3 CSV bit-for-bit (deviation $1.4e-16$) — the extended pipeline has not shifted the base. The degenerate corner $W = 0$ with the midpoint profile produces exact multiplicities and breaks the polynomial unfolding; that corner is replaced by $W = 0.1$ with an explicit note in the data table.

On the torus the flux shift is rounded to the realisable $\pm 1/24$: the constraint $\alpha \cdot N_y \in \mathbb{Z}$ on the 24-lattice is rigid, and these are exact neighbouring points of the self-dual line, not approximations.

Results: the gap is stable

The main result is negative and therefore valuable. All 35 new gates sit in the GUE class: $\langle r \rangle = 0.498\text{--}0.505$ (0.500 ± 0.002 across the family). The correlation with the zeros satisfies $|r_{H7}| \leq 0.043$ against the best protocol value 0.226 and positive controls of 1.000; the trace projection r_{EF} (C4) is distributed as 0.52 ± 0.52 — exactly inside the null clouds (surrogate, Poisson, GUE), and the decoys behave the same. No handle — alone or in combinations (anisotropy+mass, twist+mass, midpoint+torus) — moves a gate out of the null zone.

Handle by handle: twists give $|r_{H7}| \leq 0.033$ (spin structures change the global phase but not the spectral class); anisotropy ≤ 0.030 ; flux shift ≤ 0.034 ; mass ≤ 0.032 — the chiral breaking is the safest; profile ≤ 0.033 ; signs and boundaries ≤ 0.044 . The family maximum is reached on a random-sign decoy — the expected noise maximum.

Table 1. The six new handles: maxima of $|r_{H7}|$

Handle	Values	max $ r_{H7} $
Twists (φ_x , φ_y)	3 spin-structure classes	0.033
Anisotropy t_x/t_y	0.8; 1.25	0.030
Flux shift $\delta\alpha$	± 0.02 ; $\pm 1/24$ (torus)	0.034

Handle	Values	max $ r_{H7} $
Staggered mass m	0.15; 0.35	0.032
Vortex profile	site; midpoint	0.033
Signs and boundaries	neutral/all_plus/random; open/torus	0.044

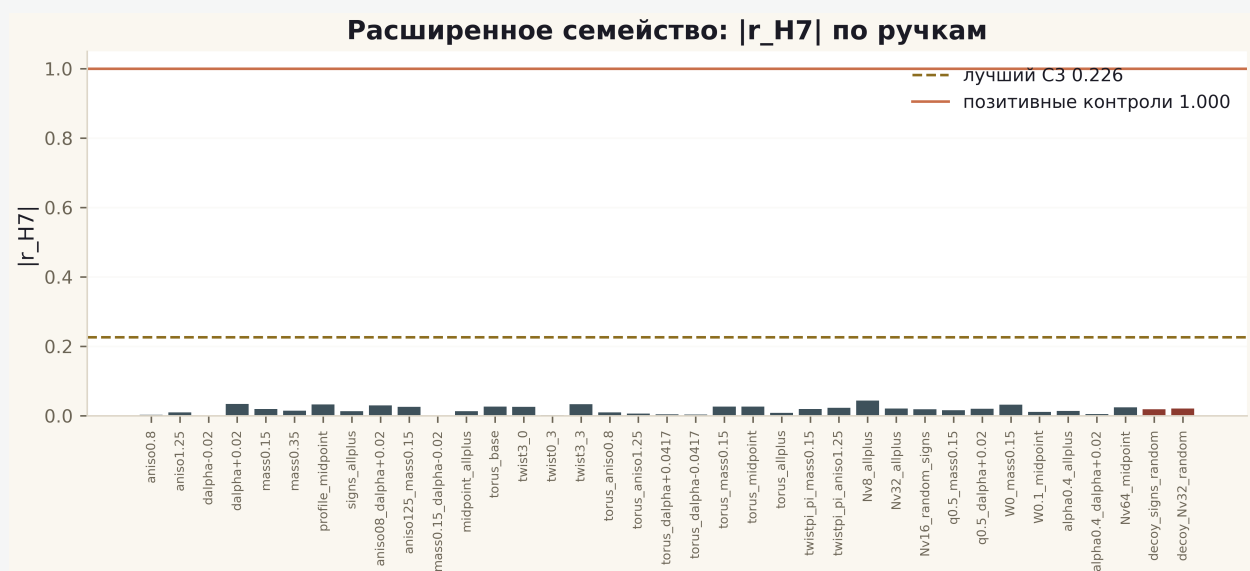


Fig. 1. $|r_{H7}|$ across the 37 extended gates: gold line — best protocol value 0.226, copper — positive controls 1.000; red bars — decoys.



Fig. 2. $\langle r \rangle$ across the extended family: every point in the 0.47–0.53 band — the GUE class is universal for all handles.

Conclusions

The H2 gap is stable against a six-dimensional extension of the handle space. Combined with C3 this means: in the class of local Hermitian Hamiltonians with vortex phases, GUE universality is an attractor that erases the arithmetic setting no matter how it is imposed. A hypothetical operator carrying the $\Lambda(n)/\sqrt{n}$ weight in its trace (T6) cannot be obtained by local deformation of the lattice resonator — a statement now supported by

87 configurations across two campaigns.